

Day 4 — General Subtraction via Complements

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: use add-the-complement trick for arbitrary subtractions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Beyond powers of 10.** Most real subtractions aren’t from a clean 10^k . So how does Nikhilam help with, say, $6347 - 2589$?
2. **The “add-the-complement” trick.** Rewrite: $6347 - 2589 = 6347 + (10000 - 2589) - 10000 = 6347 + (\text{complement of } 2589) - 10000$.
 - Complement of 2589 to 10000: $9-2=7$, $9-5=4$, $9-8=1$, $10-9=1 \rightarrow 7411$.
 - $6347 + 7411 = 13758$.
 - $13758 - 10000 = 3758$.
 - Check: $6347 - 2589 = 3758$. ✓
3. **When is this faster?** When the borrowing in standard subtraction would chain through many digits, the complement method skips it entirely. For simple subtractions, standard is fine.
4. **Worksheet.** 8 problems, mixed: some clean Nikhilam, some general using add-the-complement.
5. **Reflection.** “*Which method-type did you find genuinely faster? Which felt the same?*” Honest answers — Nikhilam is a tool, not a universal speed-up.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces General Subtraction via Complements.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).